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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Data Solutions (DATA).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Introduction

Several NGSI-LD Context Brokers and related software modules, such as data ingestion, event processing, and visualization, are being developed globally, reflecting the widespread adoption of NGSI-LD standards. With multiple open-source implementations available, establishing comprehensive interoperability test cases is essential to ensure that different brokers, even from various vendors, can operate together seamlessly. While these tests do not directly validate the NGSI-LD specification [1], these tests may also be used to demonstrate whether the implementations can interoperate effectively and whether the NGSI-LD specification should be revised to avoid ambiguity (if any). The results can be applied in specific scenarios to highlight system compatibility, showcase cross-vendor collaboration, and provide valuable feedback for improving overall implementation quality.

1 Scope

The present document extends the draft interoperability test descriptions agreed with TC DATA.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI GS CIM 009 (V1.8.1): "Context Information Management (CIM); NGSI-LD API".

2.2 Informative references

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

Not applicable.

3 Definition of terms, symbols and abbreviations

3.1 Terms

Void.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

GS	Group Specification
IOP_TP	InterOPerability Test Purpose
LD	Linked Data
NGSI	New Generation System Interfaces

SUT	System Under Test
TP	Test Purpose

4 Prerequisites and Test Configurations

4.1 Request body resources

4.1.1 Entities

4.1.1.0 Foreword

The following clauses provide a detailed description of the specific Entities that shall be employed to perform the interoperability tests outlined in clause 6 of the present document. Each clause contains the Entities of the given "type".

4.1.1.1 OffStreetParking


```
{
  "id": "urn:ngsi-ld:OffStreetParking:1",
  "type": "OffStreetParking",
  "name": {
    "type": "Property",
    "value": "Downtown One"
  },
  "availableSpotsNumber": {
    "type": "Property",
    "value": 121,
    "observedAt": "2017-07-29T12:05:02Z",
    "reliability": {
      "type": "Property",
      "value": 0.7
    }
  },
  "totalSpotsNumber": {
    "type": "Property",
    "value": 200
  },
  "location": {
    "type": "GeoProperty",
    "value": {
      "type": "Point",
      "coordinates": [
        -8.5,
        41.2
      ]
    }
  }
}
```


Figure 4.1.1.1-1: Full version of OffStreetParking1

```
{
  "id": "urn:ngsi-ld:OffStreetParking:2",
  "type": "OffStreetParking",
  "name": {
    "type": "Property",
    "value": "Downtown Two"
  },
  "availableSpotsNumber": {
    "type": "Property",
    "value": 112,
    "observedAt": "2017-07-29T12:05:02Z",
    "reliability": {
      "type": "Property",
      "value": 0.4
    }
  },
  "totalSpotsNumber": {
    "type": "Property",
    "value": 150
  },
  "location": {
    "type": "GeoProperty",
    "value": {
      "type": "Point",
      "coordinates": [
        -8.45,
        41.2
      ]
    }
  }
}
```


Figure 4.1.1.1-2: Full version of OffStreetParking2


```
{
  "id": "urn:ngsi-ld:OffStreetParking:1",
  "type": "OffStreetParking",
  "name": {
    "type": "Property",
    "value": "Downtown One"
  },
  "location": {
    "type": "GeoProperty",
    "value": {
      "type": "Point",
      "coordinates": [
        -8.9,
        42.2
      ]
    },
    "observedAt": "2021-04-04T11:45:00Z"
  }
}
```


Figure 4.1.1.1-3: OffStreetParking1 with location and name only

```
{
  "id": "urn:ngsi-ld:OffStreetParking:1",
  "type": "OffStreetParking",
  "name": {
    "type": "Property",
    "value": "Downtown One"
  },
  "availableSpotsNumber": {
    "type": "Property",
    "value": 169,
    "observedAt": "2017-07-29T12:10:02Z",
    "reliability": {
      "type": "Property",
      "value": 0.3
    }
  },
  "totalSpotsNumber": {
    "type": "Property",
    "value": 200
  }
}
```


Figure 4.1.1.1-4: OffStreetParking1 without location


```
{
  "id": "urn:ngsi-ld:OffStreetParking:2",
  "type": "OffStreetParking",
  "name": {
    "type": "Property",
    "value": "Downtown Two"
  },
  "location": {
    "type": "GeoProperty",
    "value": {
      "type": "Point",
      "coordinates": [
        -8.46,
        41.25
      ]
    },
    "observedAt": "2021-04-04T11:46:00Z"
  }
}
```


Figure 4.1.1.1-5: OffStreetParking2 with location and name only


```
{
  "id": "urn:ngsi-ld:OffStreetParking:2",
  "type": "OffStreetParking",
  "name": {
    "type": "Property",
    "value": "Downtown Two"
  },
  "availableSpotsNumber": {
    "type": "Property",
    "value": 92,
    "observedAt": "2017-07-30T12:10:02Z",
    "reliability": {
      "type": "Property",
      "value": 0.56
    }
  }
}
```


Figure 4.1.1.1-6: OffStreetParking2 without location and totalSpotsNumber

```
{
  "id": "urn:ngsi-ld:OffStreetParking:2",
  "type": "OffStreetParking",
  "name": {
    "type": "Property",
    "value": "Downtown Two"
  },
  "availableSpotsNumber": {
    "type": "Property",
    "value": 87,
    "observedAt": "2017-07-29T12:10:02Z",
    "reliability": {
      "type": "Property",
      "value": 0.55
    }
  },
  "totalSpotsNumber": {
    "type": "Property",
    "value": 150
  }
}
```

Figure 4.1.1.1-7: OffStreetParking2 without location

4.1.1.2 Vehicle

```
{
  "id": "urn:ngsi-ld:Vehicle:1",
  "type": "Vehicle",
  "brandName": {
    "type": "Property",
    "value": "Brand1"
  },
  "speed": {
    "type": "Property",
    "value": 0,
    "source": {
      "type": "Property",
      "value": "Speedometer"
    }
  },
  "isParked": {
    "type": "Relationship",
    "object": "urn:ngsi-ld:OffStreetParking:1",
    "observedAt": "2017-07-29T12:00:04Z"
  },
  "location": {
    "type": "GeoProperty",
    "value": {
      "type": "Point",
      "coordinates": [
        -8.45,
        41.2
      ]
    }
  }
}
```

Figure 4.1.1.2-1: Full version of Vehicle1

```
{
  "id": "urn:ngsi-ld:Vehicle:2",
  "type": "Vehicle",
  "brandName": {
    "type": "Property",
    "value": "Brand2"
  },
  "speed": {
    "type": "Property",
    "value": 5,
    "source": {
      "type": "Property",
      "value": "Speedometer"
    }
  },
  "isParked": {
    "type": "Relationship",
    "object": "urn:ngsi-ld:OffStreetParking:2",
    "observedAt": "2017-07-30T12:00:04Z"
  },
  "location": {
    "type": "GeoProperty",
    "value": {
      "type": "Point",
      "coordinates": [
        -9.45,
        40.2
      ]
    }
  }
}
```


Figure 4.1.1.2-2: Full version of Vehicle2

4.1.2 Attribute Fragments

None.

4.1.3 CSourceRegistrations

4.1.3.0 Foreword

The following clauses provide a detailed description of the specific CSourceRegistrations that shall be employed to perform the interoperability tests outlined in clause 6 of the present document. Each clause contains the CSourceRegistrations of the given "mode".

NOTE: In all subsequential clauses, the "endpoint" member is represented containing the placeholder text "xxx". The text should be replaced with a valid endpoint URL during the set-up of a test configuration.

4.1.3.1 Inclusive

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Inclusive:1",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "OffStreetParking"
        }
      ]
    }
  ],
  "mode": "inclusive",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.1-1: Context source registration inclusive 1

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Inclusive:2",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "OffStreetParking"
        }
      ],
      "propertyNames": [
        "availableSpotsNumber",
        "totalSpotsNumber"
      ]
    }
  ],
  "mode": "inclusive",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.1-2: Context source registration inclusive 2

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Inclusive:3",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "Vehicle"
        }
      ]
    }
  ],
  "mode": "inclusive",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.1-3: Context source registration inclusive 3

4.1.3.2 Auxiliary

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Auxiliary:1",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "OffStreetParking"
        }
      ]
    }
  ],
  "mode": "auxiliary",
  "endpoint": "xxx"

}
```

Figure 4.1.3.2-1: Context source registration auxiliary 1

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Auxiliary:2",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "OffStreetParking"
        }
      ],
      "propertyNames": [
        "availableSpotsNumber",
        "totalSpotsNumber"
      ]
    }
  ],
  "mode": "auxiliary",
  "endpoint": "xxx"
}
```

Figure 4.1.3.2-2: Context source registration auxiliary 2

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Auxiliary:3",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "Vehicle"
        }
      ]
    }
  ],
  "mode": "auxiliary",
  "endpoint": "xxx"
}
```

Figure 4.1.3.2-3: Context source registration auxiliary 3

4.1.3.3 Exclusive

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Exclusive:1",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "id": "urn:ngsi-ld:OffStreetParking:2",
          "type": "OffStreetParking"
        }
      ],
      "propertyNames": [
        "totalSpotsNumber"
      ]
    }
  ],
  "mode": "exclusive",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.3-1: Context source registration exclusive 1

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Exclusive:2",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "id": "urn:ngsi-ld:OffStreetParking:2",
          "type": "OffStreetParking"
        }
      ],
      "propertyNames": [
        "location"
      ]
    }
  ],
  "mode": "exclusive",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.3-2: Context source registration exclusive 2

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Exclusive:3",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "id": "urn:ngsi-ld:OffStreetParking:1",
          "type": "OffStreetParking"
        }
      ],
      "propertyNames": [
        "location"
      ]
    }
  ],
  "mode": "exclusive",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.3-3: Context source registration exclusive 3

4.1.3.4 Redirect

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Redirect:1",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "OffStreetParking"
        }
      ]
    }
  ],
  "mode": "redirect",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.4-1: Context source registration redirect 1

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Redirect:2",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "OffStreetParking"
        }
      ],
      "propertyNames": [
        "location"
      ]
    }
  ],
  "mode": "redirect",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.4-2: Context source registration redirect 2

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Redirect:3",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "OffStreetParking",
          "id": "urn:ngsi-ld:OffStreetParking:2"
        }
      ],
      "propertyNames": [
        "location"
      ]
    }
  ],
  "mode": "redirect",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.4-3: Context source registration redirect 3

```
{
  "id": "urn:ngsi-ld:ContextSourceRegistration:Redirect:4",
  "type": "ContextSourceRegistration",
  "information": [
    {
      "entities": [
        {
          "type": "Vehicle",
          "id": "urn:ngsi-ld:Vehicle:2"
        }
      ],
      "propertyNames": [
        "location"
      ]
    }
  ],
  "mode": "redirect",
  "operations": [
    "redirectionOps"
  ],
  "endpoint": "xxx"
}
```

Figure 4.1.3.4-4: Context source registration redirect 4

4.2 Test Configurations

4.2.0 Foreword

The following clauses provide a detailed description of the specific test configurations that shall be employed to perform the interoperability tests outlined in clause 6 of the present document. Each clause includes a diagram illustrating the topology of the configuration, a formal description that explains the rationale for selecting the given configuration, and a comprehensive list of the Context Source Registrations required when setting up the configuration.

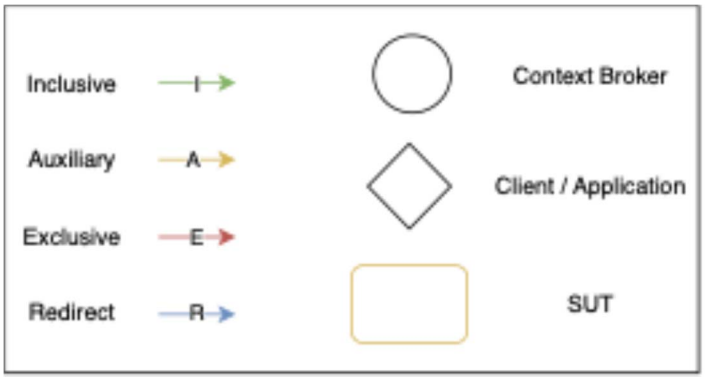


Figure 4.2.0-1: Legend

Figure 4.2.0-1 illustrates the legend for each element utilized in the diagrams of the configurations described in the present document. As depicted, registrations are represented as arcs, characterized by a colour and a label that denote a specific Context Source Registration mode. Nodes are depicted as circles. Additionally, a dashed line is employed to indicate interactions with a specific node originating from an element external to the System Under Test (SUT), typically a client.

4.2.1 IOP_CNF_01: Three brokers with Inclusive and Exclusive

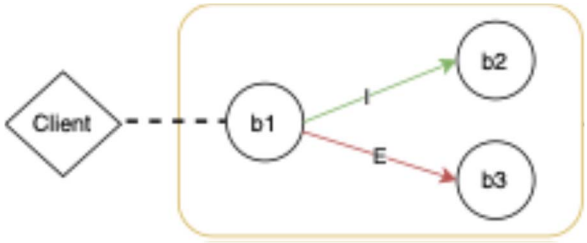


Figure 4.2.1-1: IOP_CNF_01 topology with registrations

Figure 4.2.1-1 shows the topology for IOP_CNF_01 which is one of the simplest possible setups, consisting of only three brokers. It is suitable for both Provision and Consumption operations, testing a scenario where broker b2 mirrors most of the information contained in b1 and b3 contains some exclusive data that cannot be stored in b1.

When this configuration is utilized, the following **registrations** shall be established:

- From b1 to b2, Inclusive
- From b1 to b3, Exclusive

Considering the above registrations, the following **restriction** are to be expected:

- b1 cannot store any input data that matches against the Exclusive registration

The following **behaviours** are to be expected when a **Consumption** operation is triggered on b1:

- b1 searches locally when the input data **does not** match against the Exclusive registration established with b3
- b2 is contacted by b1when the input data matches against the Inclusive registration
- b3 is the **only** broker contacted by b1 when the input data matches against the Exclusive registration

The following **behaviours** are to be expected when a **Provision** operation is triggered on b1:

- b1 writes locally when the input data **does not** match against the Exclusive registration established with b3
- b2 is contacted by b1 when the input data matches against the Inclusive registration
- b3 is the **only** broker contacted by b1 when the input data matches against the Exclusive registration.

4.2.2 IOP_CNF_02: Four brokers with Inclusive and Redirect

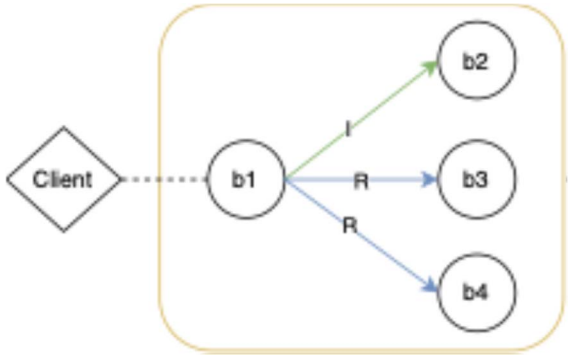


Figure 4.2.2-1: IOP_CNF_02 topology with registrations

Figure 4.2.2-1 shows the topology for IOP_CNF_02 which is a variant of IOP_CNF_01 with four brokers and Redirect registrations. It leverages the same approach, but it allows for more flexible tests. This is justified by the fact that Redirect registrations do not require to specify an Entity id, but can work with Entity types.

When this configuration is utilized, the following **registrations** shall be established:

- From b1 to b2, Inclusive
- From b1 to b3, Redirect
- From b1 to b4, Redirect

Considering the above registrations, the following **restrictions** are to be expected:

- b1 cannot store any input data that matches against the Redirect registration established with b3
- b1 cannot store any input data that matches against the Redirect registration established with b4

The following **behaviours** are to be expected when a **Consumption** operation is triggered on b1:

- b1 searches locally when the input data **does not** match against:
  - the Redirect registration established with b3
  - the Redirect registration established with b4
- b2 is contacted by b1when the input data matches against the Inclusive registration
- b3 is contacted by b1 when the input data matches against the Redirect registration
- b4 is contacted by b1 when the input data matches against the Redirect registration

The following **behaviours** are to be expected when a **Provision** operation is triggered on b1:

- b1 writes locally when the input data **does not** match against:
  - the Redirect registration established with b3
  - the Redirect registration established with b4
- b2 is contacted by b1when the input data matches against the Inclusive registration
- b3 is contacted by b1 when the input data matches against the Redirect registration
- b4 is contacted by b1 when the input data matches against the Redirect registration.

4.2.3 IOP_CNF_03: Four brokers with Auxiliary and Inclusive

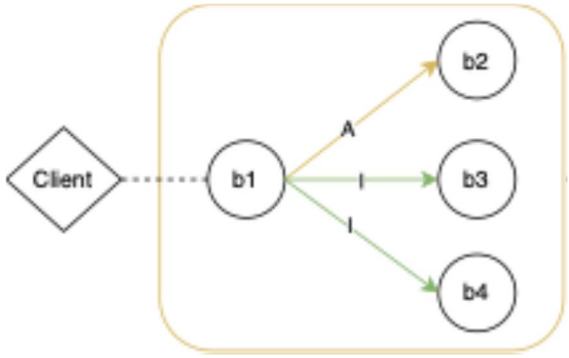


Figure 4.2.3-1: IOP_CNF_03 topology with registrations

Figure 4.2.3-1 shows the topology for IOP_CNF_03 which is a variant of IOP_CNF_02 with Auxiliary and Inclusive registrations. Considering the presence of an Auxiliary registration, this configuration is very suitable for testing Consumption operations. Provision operations can be tested as well, but the focus shifts towards a more "negative" approach where the objective is checking that an operation was **not** performed when specific requirements are met.

When this configuration is utilized, the following **registrations** shall be established:

- From b1 to b2, Auxiliary
- From b1 to b3, Inclusive
- From b1 to b4, Inclusive

The following **behaviours** are to be expected when a **Consumption** operation is triggered on b1:

- b1 searches locally
- b2 is contacted by b1 when the input data matches against the Auxiliary registration, but its data is retained **only** when all following points are met:
  - Data is not found locally within b1
  - Data is not returned by b3
  - Data is not returned by b4
- b3 is contacted by b1 when the input data matches against the Inclusive registration
- b4 is contacted by b1 when the input data matches against the Inclusive registration

The following **behaviours** are to be expected when a **Provision** operation is triggered on b1:

- b1 writes locally
- b2 is **not** contacted by b1
- b3 is contacted by b1 when the input data matches against the Inclusive registration
- b4 is contacted by b1 when the input data matches against the Inclusive registration.

4.2.4 IOP_CNF_04: Five brokers with all registration modes

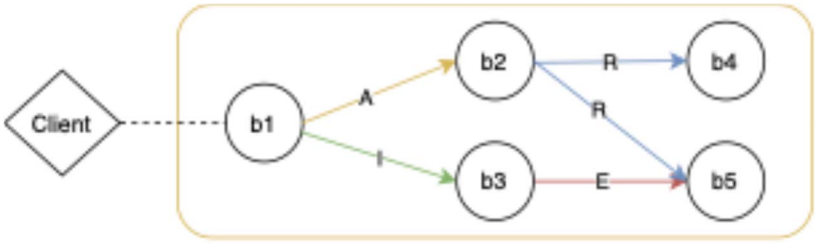


Figure 4.2.4-1: IOP_CNF_04 topology with registrations

Figure 4.2.4-1 shows the topology for IOP_CNF_04. This configuration contains all registration modes, enabling the execution of complex and comprehensive tests, particularly for evaluating Consumption operations. However, it is less suitable for Provision operations due to the presence of the Auxiliary registration from b1 to b2. In the case of a Provision request, b1 will not contact b2, which implies that nodes b2, b4, and potentially b5 (depending on the specific request) will remain unaffected.

When this configuration is utilized, the following **registrations** shall be established:

- From b1 to b2, Auxiliary
- From b2 to b4, Redirect
- From b2 to b5, Redirect
- From b1 to b3, Inclusive
- From b3 to b5, Exclusive

Considering the above registrations, the following **restrictions** are to be expected:

- b2 cannot store any input data that matches against the Redirect registration established with b4
- b2 cannot store any input data that matches against the Redirect registration established with b5
- b3 cannot store any input data that matches against the Exclusive registration established with b5

The following **behaviours** are to be expected when a **Consumption** operation is triggered on b1:

- b1 searches locally
- b2 is contacted by b1 when the input data matches against the Auxiliary registration, but its data is retained **only** when all following points are met:
  - Data is not found locally within b1
  - Data is not returned by b3
- When b2 is contacted by b1, the following behaviours are expected:
  - b2 searches locally when the input data **does not** match against:
    - the Redirect registration established with b4
    - the Redirect registration established with b5
  - b4 is contacted by b2 when the input data matches against the Redirect registration
  - b5 is contacted by b2 when the input data matches against the Redirect registration
- b3 is contacted by b1 when the input data matches against the Inclusive registration

- When b3 is contacted by b1:
  - b3 searches locally when the input data **does not** match against the Exclusive registration established with b5
  - b5 is contacted by b3 when the input data matches against the Exclusive registration

The following **behaviours** are to be expected when a **Provision** operation is triggered on b1:

- b1 writes locally
- b2 is **not** contacted by b1. Thus:
  - b4 is **not** contacted by b2
  - b5 is **not** contacted by b2
- b3 is contacted by b1 when the input data matches against the Inclusive registration. Thus:
  - b3 writes locally when the input data **does not** match against the Exclusive registration established with b5
  - b5 is contacted by b3 **when** b1 contacts b2 **and** the input data matches against the Exclusive registration.

4.3 Initial Conditions

4.3.1 Initial data

4.3.1.1 No data in any brokers

All NGSI-LD brokers shall start in a clean state with no pre-existing context data. This setup is strongly related to tests involving Provision operations (e.g. Create Entity).

4.3.1.2 Data in the local broker

Only the local broker (the one directly contacted by the client) shall start with pre-existing context data. This setup can support both Provision and Consumption operations, but it is particularly suggested for testing the merge mechanism when provisioning context data.

4.3.1.3 Data in brokers that are a leaf

Only the leaf brokers in the architecture shall start with pre-existing context data. This setup can support both Provision and Consumption operations, but it is particularly suggested for testing the merge mechanism when consuming context data.

4.3.1.4 Data in all brokers

All NGSI-LD brokers shall start with pre-existing context data. This setup is strongly related to tests involving Consumption operations (e.g. Query Entities).

4.3.2 @context file

4.3.2.1 Core context only

When this configuration is applied, user contexts shall not be included in the HTTP requests in any way. In other words, embedding a user context in either the body of the NGSI-LD payload or in the LINK header is not allowed.

4.3.2.2 User context

When this configuration is applied, user contexts shall be included in the HTTP requests. In other words, it is required to embed a user context either in the body of the NGSI-LD payload or in the Link header.

The following user context shall be used when this initial condition is included:

```
"@context": [
   {
        "Building": "https://ngsi-ld-test-suite/context#Building",
        "Vehicle": "https://ngsi-ld-test-suite/context#Vehicle",
        "OffStreetParking": "https://ngsi-ld-test-suite/context#OffStreetParking",
        "airQualityLevel": "https://ngsi-ld-test-suite/context#airQualityLevel",
        "almostFull": "https://ngsi-ld-test-suite/context#almostFull",
        "availableSpotsNumber": "https://ngsi-ld-test-suite/context#availableSpotsNumber",
        "brandName": "https://ngsi-ld-test-suite/context#brandName",
        "fuelLevel": "https://ngsi-ld-test-suite/context#fuelLevel",
        "isParked": "https://ngsi-ld-test-suite/context#isParked",
        "name": "https://ngsi-ld-test-suite/context#name",
        "reliability": "https://ngsi-ld-test-suite/context#reliability",
        "source": "https://ngsi-ld-test-suite/context#source",
        "speed": "https://ngsi-ld-test-suite/context#speed",
        "totalSpotsNumber": "https://ngsi-ld-test-suite/context#totalSpotsNumber",
        "subCategory": "https://ngsi-ld-test-suite/context#subCategory"
    },
    "https://uri.etsi.org/ngsi-ld/v1/ngsi-ld-core-context-v1.9.jsonld"
]
```

5 Interoperability Test Structure (ITS)

5.1 Test Case Structure

This clause contains the structure that shall be used for drafting the test cases included in clause 6 of the present document.

It follows the list of the Distributed Operations defined in ETSI GS CIM 009 [1] that can be chosen as an <Operation>:

- Create Entity
- Retrieve Entity
- Query Entities
- Merge Entity
- Replace Entity
- Delete Entity
- Update Entity Attribute
- Append Entity Attribute
- Partial Attribute Update
- Replace Attribute
- Delete Entity Attribute

Each <Test Case> consists of:

- A list of human-readable pre-conditions, describing the initial conditions that one shall comply to before executing the steps of a test case. Pre-conditions shall be taken from clause 4.3.
- A list of human-readable steps, describing each step that one shall follow when executing the test case. A test case is considered passed when all steps pass successfully.

Within the list of human-readable steps, the word "Agent" is used to refer either:

- A human performing and evaluating the test manually
- A software performing and evaluating the test programmatically.

6 Test Descriptions

6.0 Foreword

The following clauses provide a detailed description of each test case.

Each subclause is structured as it follows:

- <Operation>
  - <Test Configuration>
    - <Test Case>

Where:

- <Operation> is the name of the specific Distributed Operation to test.
- <Test Configuration> is the code of a test configuration taken from clause 4.2.
- <Test Case> is a brief description of the end-goal of the test case.

6.1 CreateEntity

6.1.1 IOP_CNF_01

6.1.1.1 Create OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_01_01
Test objective	Three brokers are set up b1, b2 and b3. b1 has two registrations, one inclusive for the entities in b2 and one exclusive for the entity in b3. Check that the entity is fully created in b1. Check that the entity is created in b2 and only contains the availableSpotsNumber and totalSpotsNumber attributes. Check that the entity is not created in b3.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.6.1
Config Id	IOP_CNF_01
PICS Selection	PICS_DISTRIBUTED and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** No data in any broker.
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Exclusive b1 to b3 (Figure 4.1.3.3-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_01_01
Name	IOP_001_01_01 Create OffStreetParking:1
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to create the entity Full version of OffStreetParking1 (Figure 4.1.1.1-1) **2:** Agent checks that a success response has been returned **3:** Agent checks (with local=true) that Full version of OffStreetParking1 (Figure 4.1.1.1-1) was created in b1 **4:** Agent checks (with local=true) that the Entity was created in b2 and that the Entity only contains "availableSpotsNumber" and "totalSpotsNumber" **5:** Agent checks (with local=true) that the Entity was NOT created in b3

6.1.1.2 Create OffStreetParking:2

TP Id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_01_02
Test objective	Three brokers are set up b1, b2 and b3. b1 has two registrations, one inclusive for the entities in b2 and one exclusive for the entity in b3. Check that the entity is created in b1 and b2 not containing the location attribute. Check that the entity is created in b3 containing only the location attribute.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.6.1
Config Id	IOP_CNF_01
PICS Selection	PICS_DISTRIBUTED and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** No data in any broker.
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Exclusive b1 to b3 (Figure 4.1.3.3-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_01_02
Name	IOP_001_01_02 Create OffStreetParking:2
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to create the entity Full version of OffStreetParking2 (Figure 4.1.1.1-2) **2:** Agent checks that a success response has been returned **3:** Agent checks (with local=true) that the Entity was created in b1 and that the Entity does not contain the "location" Property **4:** Agent checks (with local=true) that the Entity was created in b2 and that the Entity does not contain the "location" Property **5:** Agent checks (with local=true) that the Entity was created in b3 and that the Entity contains only the "location" Property

6.1.2 IOP_CNF_02

6.1.2.1 Create OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_02_01
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one inclusive for the entities in b2, one redirect for the entity in b3 and one redirect for the entities in b4. Check that the entity is created in b1 and it does not contain the location property. Check that the entity is created in b2 and does not contain the location and name attributes. Check that the entity is created in b3 and b4 containing only the location attribute.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.6.1
Config Id	IOP_CNF_02
PICS Selection	PICS_DISTRIBUTED and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** No data in any broker.
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Redirect b1 to b3 (Figure 4.1.3.4-2). • Redirect b1 to b4 (Figure 4.1.3.4-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_02_01
Name	IOP_001_02_01 Create OffStreetParking:1
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to create the entity Full version of OffStreetParking1 (Figure 4.1.1.1-1) **2:** Agent checks that a success response has been returned **3:** Agent checks (with local=true) that the Entity was created in b1 and that the Entity does not contain the "location" Property **4:** Agent checks (with local=true) that the Entity was created in b2 and that the Entity does not contain the "location" and "name" Properties **5:** Agent checks (with local=true) that the Entity was created in b3 and that the Entity contains only the "location" Property **6:** Agent checks (with local=true) that the Entity was created in b4 and that the Entity contains only the "location" Property

6.1.2.2 Create OffStreetParking:2

TP Id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_02_02
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one inclusive for the entities in b2, one redirect for the entity in b3 and one redirect for the entities in b4. Check that the entity is created in b1 and does not contain the location attribute. Check that the entity is created in b2 and does not contain the location and name properties. Check that the entity is created in b3 and b4 containing only the location attribute.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.6.1
Config Id	IOP_CNF_02
PICS Selection	PICS_DISTRIBUTED and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** No data in any broker.
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Redirect b1 to b3 (Figure 4.1.3.4-2). • Redirect b1 to b4 (Figure 4.1.3.4-3).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_02_02
Name	IOP_001_02_02 Create OffStreetParking:2
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to create the entity Full version of OffStreetParking2 (Figure 4.1.1.1-2) **2:** Agent checks that a success response has been returned **3:** Agent checks (with local=true) that the Entity was created in b1 and that the Entity does not contain the "location" Property **4:** Agent checks (with local=true) that the Entity was created in b2 and that the Entity does not contain the "location" and "name" Properties **5:** Agent checks (with local=true) that the Entity was created in b3 and that the Entity contains only the "location" Property **6:** Agent checks (with local=true) that the Entity was created in b4 and that the Entity contains only the "location" Property

6.1.3 IOP_CNF_03

6.1.3.1 Create OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_03_01
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one auxiliary for the entities in b2, one inclusive for the entity in b3 and one inclusive for the entities in b4. Check that the entity is created in b1 and b3. The entity shall not be present in b3. The entity in b4 shall only contain availableSpotsNumber and totalSpotsNumber.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.6.1
Config Id	IOP_CNF_03
PICS Selection	PICS_DISTRIBUTED and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** No data in any broker.
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-2). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Inclusive b1 to b4 (Figure 4.1.3.1-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_03_01
Name	IOP_001_03_01 Create OffStreetParking:1
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to create the entity Full version of OffStreetParking1 (Figure 4.1.1.1-1) **2:** Agent checks that a success response has been returned **3:** Agent checks (with local=true) that the Entity was created in b1 **4:** Agent checks (with local=true) that the Entity was NOT created in b2 **5:** Agent checks (with local=true) that the Entity was created in b3 **6:** Agent checks (with local=true) that the Entity was created in b4 and that it only contains "availableSpotsNumber" and "totalSpotsNumber"

6.1.3.2 Create Partial OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_03_02
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one auxiliary for the entities in b2, one inclusive for the entity in b3 and one inclusive for the entities in b4. Check that the entity is created in b1 and b3. The entity shall not be present in b2 and b4.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.6.1
Config Id	IOP_CNF_03
PICS Selection	PICS_DISTRIBUTED and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** No data in any broker.
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-2). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Inclusive b1 to b4 (Figure 4.1.3.1-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_03_02
Name	IOP_001_03_02 Create Partial OffStreetParking:1
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to create the entity OffStreetParking1 with location and name only (Figure 4.1.1.1-3) **2:** Agent checks that a success response has been returned **3:** Agent checks (with local=true) that the Entity was created in b1 **4:** Agent checks (with local=true) that the Entity was NOT created in b2 **5:** Agent checks (with local=true) that the Entity was created in b3 **6:** Agent checks (with local=true) that the Entity was NOT created in b4

6.1.4 IOP_CNF_04

6.1.4.1 Create OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_04_01
Test objective	Five brokers are set up b1, b2, b3, b4 and b5. b1 has two registrations, one auxiliary for the entities in b2 and one inclusive for the entity in b3. b2 has two registration, one redirect for the entities in b4, one redirect for the entity in b5. b3 has one exclusive registration to b5. Check that the entity is created in b1 and b3. The entity shall not be created in b2, b4 and b5.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.6.1
Config Id	IOP_CNF_04
PICS Selection	PICS_DISTRIBUTED and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** No data in any broker.
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-2). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Redirect b2 to b4 (Figure 4.1.3.4-1). • Redirect b2 to b5 (Figure 4.1.3.4-2). • Exclusive b3 to b5 (Figure 4.1.3.3-1).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_04_01
Name	IOP_001_04_01 Create OffStreetParking:1
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to create the entity Full version of OffStreetParking1 (Figure 4.1.1.1-1) **2:** Agent checks that a success response has been returned **3:** Agent checks (with local=true) that the Entity was created in b1 **4:** Agent checks (with local=true) that the Entity was NOT created in b2 **5:** Agent checks (with local=true) that the Entity was created in b3 **6:** Agent checks (with local=true) that the Entity was NOT created in b4 **7:** Agent checks (with local=true) that the Entity was NOT created in b5

6.1.4.2 Create OffStreetParking:2

TP Id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_04_02
Test objective	Five brokers are set up b1, b2, b3, b4 and b5. b1 has two registrations, one auxiliary for the entities created in b2 and one inclusive for the entity created in b3. b2 has two registration, one redirect for the entities created in b4, one redirect for the entity created in b5. b3 has one exclusive registration to b5. Check that the entity is created in b1 and b3. The entity shall not be created in b2 and b4. The entity shall be created in b5 with only one the totalSpotsNumber attribute.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.6.1
Config Id	IOP_CNF_04
PICS Selection	PICS_DISTRIBUTED and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** No data in any broker.
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-2). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Redirect b2 to b4 (Figure 4.1.3.4-1). • Redirect b2 to b5 (Figure 4.1.3.4-2). • Exclusive b3 to b5 (Figure 4.1.3.3-1).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Prov/CreateEntity/IOP_001_04_02
Name	IOP_001_04_02 Create OffStreetParking:2
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to create the entity Full version of OffStreetParking2 (Figure 4.1.1.1-2) **2:** Agent checks that a success response has been returned **3:** Agent checks (with local=true) that the Entity was created in b1 **4:** Agent checks (with local=true) that the Entity was NOT created in b2 **5:** Agent checks (with local=true) that the Entity was created in b3 **6:** Agent checks (with local=true) that the Entity was NOT created in b4 **7:** Agent checks (with local=true) that the Entity was created in b5 and that the Entity only contains the "totalSpotsNumber" Property

6.2 RetrieveEntity

6.2.1 IOP_CNF_01

6.2.1.1 Retrieve OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_01_01
Test objective	Three brokers are set up b1, b2 and b3. b1 has two registrations, one inclusive for the entities in b2 and one exclusive for the entity in b3. Check that the OffStreetParking:1 entity returned from b2 matches the full OffStreetParking:1 entity.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.1
Config Id	IOP_CNF_01
PICS Selection	PICS_DISTRIBUTED and PICS_EntRet
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b2 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2). – b3 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2).
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Exclusive b1 to b3 (Figure 4.1.3.3-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_01_01
Name	IOP_002_01_01 Retrieve OffStreetParking:1
Parent Release	v1.6.1

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_01_01
Expected behaviour	**Test events**
	1: Client sends an HTTP GET request to b1 to retrieve the OffStreetParking:1 **2:** Agent checks that a success response has been returned and that only availableSpotsNumber and totalSpotsNumber of OffStreetParking:1 were returned **3:** Client sends an HTTP GET request to b2 to retrieve the OffStreetParking:1 **4:** Agent checks that the Entity returned in step (3) matches Full version of OffStreetParking1 (Figure 4.1.1.1-1)

6.2.1.2 Retrieve OffStreetParking:2

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_01_02
Test objective	Three brokers are set up b1, b2 and b3. b1 has two registrations, one inclusive for the entity in b2 and one exclusive for the entity in b3. Check that the OffStreetParking:2 entity returned from b1 has the same availableSpotsNumber and totalSpotsNumber attributes as the ones returned from b2. Check that the OffStreetParking:2 entity returned from b1 has the same location attribute as the one returned from b3.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.1
Config Id	IOP_CNF_01
PICS Selection	PICS_DISTRIBUTED and PICS_EntRet
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains OffStreetParking2 without location (Figure 4.1.1.1-7). – b3 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2).
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Exclusive b1 to b3 (Figure 4.1.3.3-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_01_02
Name	IOP_002_01_02 Retrieve OffStreetParking:2
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP GET request to b1 to retrieve the Entity OffStreetParking:2 **2:** Agent checks that a success response has been returned and that the Entity OffStreetParking:2 was returned **3:** Client sends an HTTP GET request to b2 to retrieve the Entity OffStreetParking:2 **4:** Client sends an HTTP GET request to b3 to retrieve the Entity OffStreetParking:2 **5:** Agent checks that the Entity returned in step (2) is structured as follows: • The attributes "availableSpotsNumber" and "totalSpotsNumber" match the ones from the Entity returned in step (3) • The attribute "location" matches the one from the Entity returned in step (4)

6.2.2 IOP_CNF_02

6.2.2.1 Retrieve OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_02_01
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one inclusive for the entity in b2, one redirect for the entity in b3 and one redirect for the entity in b4. Check that the OffStreetParking:1 entity returned from b1 match the availableSpotsNumber and totalSpotsNumber from the entity in b2. Check that the OffStreetParking:1 entity returned from b1 match the location attribute from the entity in b3.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.1
Config Id	IOP_CNF_02
PICS Selection	PICS_DISTRIBUTED and PICS_EntRet
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b3 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b4 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2).
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Redirect b1 to b3 (Figure 4.1.3.4-2). • Redirect b1 to b4 (Figure 4.1.3.4-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_02_01

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_02_01
Name	IOP_002_02_01 Retrieve OffStreetParking:1
Parent Release	v1.6.1
Expected behaviour	Test events
	1: Client sends an HTTP GET request to b1 to retrieve the Entity OffStreetParking:1 **2:** Agent checks that a partial success response has been returned and that Full version of OffStreetParking1 (Figure 4.1.1.1-1) was returned **3:** Client sends an HTTP GET request to b2 to retrieve the Entity OffStreetParking:1 **4:** Client sends an HTTP GET request to b3 to retrieve the Entity OffStreetParking:1 **5:** Agent checks that the Entity returned in step (2) is structured as follows: • The attributes "availableSpotsNumber" and "totalSpotsNumber" match the ones from the Entity returned in step (3) • The attribute "location" matches the one from the Entity returned in step (4)

6.2.2.2 Retrieve OffStreetParking:1 Location Attribute

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_02_02
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one inclusive for the entity in b2, one redirect for the entity in b3 and one redirect for the entity in b4. Check that the OffStreetParking:1 entity returned from b1 has location attribute. Check that the location attribute from the entity returned from b1 is the same as the one returned from b3.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.1
Config Id	IOP_CNF_02
PICS Selection	PICS_DISTRIBUTED and PICS_EntRet
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b3 contains OffStreetParking1 with location and name only (Figure 4.1.1.1-3). – b4 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2).
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Redirect b1 to b3 (Figure 4.1.3.4-2). • Redirect b1 to b4 (Figure 4.1.3.4-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_02_02
Name	IOP_002_02_02 Retrieve OffStreetParking:1 Location Attribute
Parent Release	v1.6.1
Expected behaviour	Test events
	1: Client sends an HTTP GET request to b1 to retrieve the "location" Property from the Entity OffStreetParking:1 **2:** Agent checks that a partial success response has been returned and that OffStreetParking:1 with the "location" Property was returned **3:** Client sends an HTTP GET request (with local=true) to b3 to retrieve the Entity OffStreetParking:1 **4:** Agent checks that the Entity returned in step (2) contains the attribute "location" and that it matches the one from the Entity returned in step (3)

6.2.3 IOP_CNF_03

6.2.3.1 Retrieve OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_03_01
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one auxiliary for the entity in b2, one inclusive for the entity in b3 and one inclusive for the entity in b4. Check that the OffStreetParking:1 entity returned from b1 does not contain the location attribute. Check that the entity returned from b1 does not contain the location attribute returned from b3.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.1
Config Id	IOP_CNF_03
PICS Selection	PICS_DISTRIBUTED and PICS_EntRet

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_03_01
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b1 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b2 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b3 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b4 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2).
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-2). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Inclusive b1 to b4 (Figure 4.1.3.1-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_03_01
Name	IOP_002_03_01 Retrieve OffStreetParking:1
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP GET request to b1 to retrieve the Entity OffStreetParking:1 **2:** Agent checks that a partial success response has been returned and that the Entity OffstreetParking:1 without the "location" Property was returned **3:** Client sends an HTTP GET request to b1 (with local=true) to retrieve the Entity OffStreetParking:1 **4:** Client sends an HTTP GET request to b2 (with local=true) to retrieve the Entity OffStreetParking:1 **5:** Client sends an HTTP GET request to b3 (with local=true) to retrieve the Entity OffStreetParking:1 **6:** Agent checks that the Entity returned in step (2) matches the one returned in step (3) and that it does not contain the "location" Property returned in step (4)

6.2.3.2 Retrieve OffStreetParking:1 Location Attribute

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_03_02
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one auxiliary for the entity in b2, one inclusive for the entity in b3 and one inclusive for the entity in b4. Check that the entity returned from b1 has the same location attribute as the one found in b3.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.1
Config Id	IOP_CNF_03
PICS Selection	PICS_DISTRIBUTED and PICS_EntRet
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b1 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b2 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b3 contains OffStreetParking1 with location and name only (Figure 4.1.1.1-3). – b4 contains OffStreetParking1 without location (Figure 4.1.1.1-4).
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-1). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Inclusive b1 to b4 (Figure 4.1.3.1-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_03_02
Name	IOP_002_03_02 Retrieve OffStreetParking:1 Location Attribute
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP GET request to b1 to retrieve the Entity OffStreetParking:1 **2:** Agent checks that a partial success response has been returned and that the Entity OffstreetParking:1 containing only the "location" Property was returned **3:** Client sends an HTTP GET request to b3 (with local=true) to retrieve the Entity OffStreetParking:1 **4:** Agent checks that the Entity returned in step (2) contains the attribute "location" and that it matches the one from the Entity returned in step (3)

6.2.4 IOP_CNF_04

6.2.4.1 Retrieve OffStreetParking:1

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_04_01
Test objective	Five brokers are set up b1, b2, b3, b4 and b5. b1 has two registrations, one auxiliary for the entity in b2, one inclusive for the entity in b3. b2 has two registrations, one redirect for the entity in b4 and one redirect for the entity in b5. b3 shall establish one exclusive registration to b5. Check that the entity returned from b1 has the same availableSpotsNumber and totalSpotsNumber as the ones found in b4. Check that the entity returned from b1 has the same location attribute as the one in b5.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.1
Config Id	IOP_CNF_04
PICS Selection	PICS_DISTRIBUTED and PICS_EntRet
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b4 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b5 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1).
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-2). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Redirect b2 to b4 (Figure 4.1.3.4-1). • Redirect b2 to b5 (Figure 4.1.3.4-2). • Exclusive b3 to b5 (Figure 4.1.3.3-1).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_04_01
Name	IOP_002_04_01 Retrieve OffStreetParking:1
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP GET request to b1 to retrieve the Entity OffStreetParking:1 **2:** Agent checks that a success response has been returned and that the Entity OffStreetParking:1 was returned **3:** Client sends an HTTP GET request to b4 to retrieve the Entity OffStreetParking:1 **4:** Client sends an HTTP GET request to b5 to retrieve the Entity OffStreetParking:1 **5:** Agent checks that the Entity returned in step (2) is structured as follows: • The attributes "availableSpotsNumber" and "totalSpotsNumber" match the ones from the Entity returned in step (3) • The attribute "location" matches the one from the Entity returned in step (4)

6.2.4.2 Retrieve OffStreetParking:1 Location Attribute

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_04_02
Test objective	Five brokers are set up b1, b2, b3, b4 and b5. b1 has two registrations, one auxiliary for the entity in b2, one inclusive for the entity in b3. b2 has two registrations, one redirect for the entity in b4 and one redirect for the entity in b5. b3 shall establish one exclusive registration to b5. Check that the entity returned from b1 has the same availableSpotsNumber and totalSpotsNumber as the ones found in b2. Check that the entity returned from b1 has the same name attribute as the one in b3. Check that the entity returned from b1 has the same location attribute as the one in b5.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.1
Config Id	IOP_CNF_04
PICS Selection	PICS_DISTRIBUTED and PICS_EntRet
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b1 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b2 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b3 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b4 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b5 contains OffStreetParking1 with location and name only (Figure 4.1.1.1-3).
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-1). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Redirect b2 to b4 (Figure 4.1.3.4-2). • Redirect b2 to b5 (Figure 4.1.3.4-3). • Exclusive b3 to b5 (Figure 4.1.3.3-3).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_04_02

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_002_04_02
Name	IOP_002_04_02 Retrieve OffStreetParking:1 Location Attribute
Parent Release	v1.6.1
Expected behaviour	Test events
	1: Client sends an HTTP GET request to b1 to retrieve the Entity OffStreetParking:1 **2:** Agent checks that a success response has been returned and that the Entity OffStreetParking:1 with the "location" Property was returned **3:** Client sends an HTTP GET request to b2 (with local=true) to retrieve the Entity OffStreetParking:1 **4:** Client sends an HTTP GET request to b3 (with local=true) to retrieve the Entity OffStreetParking:1 **5:** Client sends an HTTP GET request to b5 (with local=true) to retrieve the Entity OffStreetParking:1 **6:** Agent checks that the Entity returned in step (2) is structured as follows: • The attributes "availableSpotsNumber" and "totalSpotsNumber" match the ones from the Entity returned in step (3) • The attribute "name" matches the one from the Entity returned in step (4) • The attribute "location" matches the one from the Entity returned in step (5)

6.3 QueryEntities

6.3.1 IOP_CNF_01

6.3.1.1 Query Entities Of Type OffstreetParking Via GET

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_01_01
Test objective	Three brokers are set up b1, b2 and b3. b1 has two registrations, one inclusive for the entities in b2 and one exclusive for the entity in b3. Check that OffStreetParking:1 in b1 contains availableSpotsNumber and totalSpotsNumber from b2. Check that OffStreetParking:2 in b1 contains availableSpotsNumber from b2 and location from b3.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.2, 6.4.3.1
Config Id	IOP_CNF_01
PICS Selection	PICS_DISTRIBUTED and PICS_QueEnt_AND_QueEntPos and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b2 contains OffStreetParking2 without location and totalSpotsNumber (Figure 4.1.1.1-6). – b3 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2).
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Exclusive b1 to b3 (Figure 4.1.3.3-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_01_01
Name	IOP_003_01_01 Query Entities Of Type OffstreetParking Via GET
Parent Release	v1.6.1
Expected behaviour	Test events
	1: Client sends an HTTP GET request to b1 to retrieve all Entities with type OffstreetParking **2:** Agent checks that a success response has been returned and that the following Entities were returned: • OffstreetParking:1 with the attributes availableSpotsNumber and totalSpotsNumber • OffstreetParking:2 with the attributes availableSpotsNumber and location **3:** Client sends an HTTP GET request to b2 to retrieve all Entities with type OffstreetParking **4:** Client sends an HTTP GET request to b3 to retrieve all Entities with type OffstreetParking **5:** Agent checks that b2 returns OffstreetParking:1 and OffstreetParking:2 and b3 returns OffstreetParking:2 **6:** Agent checks that OffstreetParking:1 returned in step (2) matches b2 and OffstreetParking:2 combines b2 and b3

6.3.1.2 Query Entities Of Type OffstreetParking Via POST

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_01_02
Test objective	Three brokers are set up b1, b2 and b3. b1 has two registrations, one inclusive for the entities in b2 and one exclusive for the entity in b3. Check that OffStreetParking:1 in b1 contains availableSpotsNumber and totalSpotsNumber from b2. Check that OffStreetParking:2 in b1 contains availableSpotsNumber from b2 and location from b3.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.2, 6.23.2.1
Config Id	IOP_CNF_01
PICS Selection	PICS_DISTRIBUTED and PICS_QueEnt_AND_QueEntPos
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b2 contains OffStreetParking2 without location and totalSpotsNumber (Figure 4.1.1.1-6). – b3 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2).
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Exclusive b1 to b3 (Figure 4.1.3.3-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_01_02
Name	IOP_003_01_02 Query Entities Of Type OffstreetParking Via POST
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to retrieve all Entities with type OffstreetParking **2:** Agent checks that a success response has been returned and that the following Entities were returned: • OffstreetParking:1 with the attributes availableSpotsNumber and totalSpotsNumber • OffstreetParking:2 with the attributes availableSpotsNumber and location **3:** Client sends an HTTP POST request to b2 to retrieve all Entities with type OffstreetParking **4:** Client sends an HTTP POST request to b3 to retrieve all Entities with type OffstreetParking **5:** Agent checks that b2 returns OffstreetParking:1 and OffstreetParking:2 and b3 returns OffstreetParking:2 **6:** Agent checks that OffstreetParking:1 returned in step (2) matches b2 and OffstreetParking:2 combines b2 and b3

6.3.2 IOP_CNF_02

6.3.2.1 Query Entities Of Type OffstreetParking Via GET

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_02_01
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one inclusive for the entities in b2, one redirect for the entities in b3 and b4 and one redirect for the entity in b4. Check that the OffStreetParking:1 entity in b1 has the same availableSpotsNumber and totalSpotsNumber attributes as the one in b2. Check that the OffStreetParking:1 entity in b1 has the same location attribute as the one found in b3. Check that the OffStreetParking:2 entity in b1 has the same availableSpotsNumber and totalSpotsNumber attributes as the one in b3. Check that the OffStreetParking:2 entity in b1 has the same location attribute as the one in b4.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.2, 6.4.3.1
Config Id	IOP_CNF_02
PICS Selection	PICS_DISTRIBUTED and PICS_QueEnt_AND_QueEntPos and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains OffStreetParking1 without location (Figure 4.1.1.1-4). – b2 contains OffStreetParking2 without location (Figure 4.1.1.1-7). – b3 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b3 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2). – b4 contains OffStreetParking2 with location and name only (Figure 4.1.1.1-5).
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Redirect b1 to b3 (Figure 4.1.3.4-2). • Redirect b1 to b4 (Figure 4.1.3.4-3).

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_02_01
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_02_01
Name	IOP_003_02_01 Query Entities Of Type OffstreetParking Via GET
Parent Release	v1.6.1
Expected behaviour	Test events
	1: Client sends an HTTP GET request to b1 to retrieve Entities with type OffstreetParking **2:** Agent checks that a success response has been returned and that OffstreetParking:1 and OffstreetParking:2 were returned with all properties but "name" **3:** Client sends an HTTP GET request to b2 to retrieve Entities with type OffstreetParking **4:** Client sends an HTTP GET request to b3 to retrieve Entities with type OffstreetParking **5:** Client sends an HTTP GET request to b4 to retrieve Entities with type OffstreetParking **6:** Agent checks the broker-b1 Entities against the attributes returned by b2, b3 and b4

6.3.2.2 Query Entities Of Type OffstreetParking And Vehicle With Attrs

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_02_02
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has five registrations, two inclusive for the entities in b2, one redirect for the entities in b3 and two redirect for the entity in b4. Check that the Vehicle:1 entity in b1 is the same as the one in b2. Check that the entity OffStreetParking:1 in b1 is the same as the one in b3. Check that the entities Vehicle:2 and OffStreetParking:2 in b1 are the same as the ones in b4.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.2, 6.4.3.1
Config Id	IOP_CNF_02
PICS Selection	PICS_DISTRIBUTED and PICS_QueEnt_AND_QueEntPos and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains Full version of Vehicle1 (Figure 4.1.1.2-1). – b2 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b3 contains OffStreetParking1 with location and name only (Figure 4.1.1.1-3). – b3 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2). – b4 contains OffStreetParking2 with location and name only (Figure 4.1.1.1-5). – b4 contains Full version of Vehicle2 (Figure 4.1.1.2-2).
Registrations Established	• Inclusive b1 to b2 (Figure 4.1.3.1-2). • Inclusive b1 to b2 (Figure 4.1.3.1-3). • Redirect b1 to b3 (Figure 4.1.3.4-2). • Redirect b1 to b4 (Figure 4.1.3.4-3). • Redirect b1 to b4 (Figure 4.1.3.4-4).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_02_02
Name	IOP_003_02_02 Query Entities Of Type OffstreetParking And Vehicle With Attrs
Parent Release	v1.6.1
Expected behaviour	Test events
	1: Client sends an HTTP GET request to b1 to retrieve the "location" Property from Entities having type OffstreetParking or Vehicle **2:** Agent checks that a success response has been returned and that all four Entities were returned with the "location" Property only **3:** Client sends an HTTP GET request to b2 to retrieve the "location" Property from Entities having type OffstreetParking or Vehicle **4:** Client sends an HTTP GET request to b3 to retrieve the "location" Property from Entities having type OffstreetParking or Vehicle **5:** Client sends an HTTP GET request to b4 to retrieve the "location" Property from Entities having type OffstreetParking or Vehicle **6:** Agent checks that Vehicle:1 matches b2, OffstreetParking:1 matches b3, and Vehicle:2 and OffstreetParking:2 match b4

6.3.3 IOP_CNF_03

6.3.3.1 Query Entities Of Type OffstreetParking Via GET

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_03_01
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one auxiliary for the entities in b2, one inclusive for the entities in b3 and one inclusive for the entity in b4. Check that the OffStreetParking:1 entity in b1 has the same name and location attributes as the ones in b3. Check that the entity OffStreetParking:1 in b1 has the same availableSpotsNumber and totalSpotsNumber attributes as the one in b4. Check that the OffStreetParking:2 entity is the same as the one in b2.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.2, 6.4.3.1
Config Id	IOP_CNF_03
PICS Selection	PICS_DISTRIBUTED and PICS_QueEnt_AND_QueEntPos and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains OffStreetParking2 without location (Figure 4.1.1.1-7). – b3 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b4 contains OffStreetParking1 without location (Figure 4.1.1.1-4).
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-1). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Inclusive b1 to b4 (Figure 4.1.3.1-2).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_03_01
Name	IOP_003_03_01 Query Entities Of Type OffstreetParking Via GET
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP GET request to b1 to retrieve Entities with type OffstreetParking **2:** Agent checks that a success response has been returned with Full version of OffStreetParking1 (Figure 4.1.1.1-1) and OffStreetParking2 without location (Figure 4.1.1.1-7) **3:** Client sends an HTTP GET request to b2 to retrieve Entities with type OffstreetParking **4:** Client sends an HTTP GET request to b3 to retrieve Entities with type OffstreetParking **5:** Client sends an HTTP GET request to b4 to retrieve Entities with type OffstreetParking **6:** Agent checks that OffstreetParking:1 combines b3 and b4 and OffstreetParking:2 matches b2

6.3.3.2 Query Entities Of Type OffstreetParking And Vehicle With Attrs

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_03_02
Test objective	Four brokers are set up b1, b2, b3 and b4. b1 has three registrations, one auxiliary for the entities in b2, one inclusive for the entities in b3 and one inclusive for the entity in b4. Check that the OffStreetParking:1 entity in b1 has the same name and location attributes as the ones in b3. Check that the OffStreetParking:1 entity in b1 has the same availableSpotsNumber and totalSpotsNumber attributes as the one in b4. Check that the OffStreetParking:2 entity is the same as the one in b4. Check that the Vehicle:1 entity in b1 is the same as the one in b2.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.2, 6.4.3.1
Config Id	IOP_CNF_03
PICS Selection	PICS_DISTRIBUTED and PICS_QueEnt_AND_QueEntPos and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b2 contains Full version of Vehicle1 (Figure 4.1.1.2-1). – b3 contains OffStreetParking1 with location and name only (Figure 4.1.1.1-3). – b4 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b4 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2). – b4 contains Full version of Vehicle2 (Figure 4.1.1.2-2).
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-3). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Inclusive b1 to b4 (Figure 4.1.3.1-1).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_03_02
Name	IOP_003_03_02 Query Entities Of Type OffstreetParking And Vehicle With Attrs
Parent Release	v1.6.1

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_03_02
Expected behaviour	**Test events**
	1: Client sends an HTTP GET request to b1 to retrieve the "location" Property from all Entities having type OffstreetParking or Vehicle **2:** Agent checks that a success response has been returned with OffstreetParking:1, OffstreetParking:2 and Vehicle:1 **3:** Client sends an HTTP GET request to b2 to retrieve the "location" Property from Entities having type OffstreetParking or Vehicle **4:** Client sends an HTTP GET request to b3 to retrieve the "location" Property from Entities having type OffstreetParking or Vehicle **5:** Client sends an HTTP GET request to b4 to retrieve the "location" Property from Entities having type OffstreetParking or Vehicle **6:** Agent checks that OffstreetParking:1 matches b3, OffstreetParking:2 matches b4 and Vehicle:1 comes from b2

6.3.4 IOP_CNF_04

6.3.4.1 Query Entities Of Type OffstreetParking Via GET

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_04_01
Test objective	Five brokers are set up b1, b2, b3, b4 and b5. b1 has two registrations, one auxiliary for the entity in b2, one inclusive for the entity in b3. b2 has two registrations, one redirect for the entity in b4 and one redirect for the entity in b5. b3 has two exclusive registrations to b5. Check that the OffStreetParking:1 entity has the same location attribute as the one in b5. Check that the OffStreetParking:2 entity in b1 has the same availableSpotsNumber and totalSpotsNumber as the ones found in b4. Check that the OffStreetParking:2 entity in b1 has the same location attribute as the one in b5.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.2, 6.4.3.1
Config Id	IOP_CNF_04
PICS Selection	PICS_DISTRIBUTED and PICS_QueEnt_AND_QueEntPos and PICS_EntCre
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b4 contains OffStreetParking1 with location and name only (Figure 4.1.1.1-3). – b4 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2). – b5 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b5 contains OffStreetParking2 with location and name only (Figure 4.1.1.1-5).
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-2). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Redirect b2 to b4 (Figure 4.1.3.4-1). • Redirect b2 to b5 (Figure 4.1.3.4-2). • Exclusive b3 to b5 (Figure 4.1.3.3-2). • Exclusive b3 to b5 (Figure 4.1.3.3-3).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_04_01
Name	IOP_003_04_01 Query Entities Of Type OffstreetParking Via GET
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP GET request to b1 to retrieve Entities with type OffstreetParking **2:** Agent checks that a success response has been returned with OffstreetParking:1 containing "location" and OffstreetParking:2 containing "availableSpotsNumber", "totalSpotsNumber" and "location" **3:** Client sends an HTTP GET request to b4 to retrieve Entities with type OffstreetParking **4:** Client sends an HTTP GET request to b5 to retrieve Entities with type OffstreetParking **5:** Agent checks that OffstreetParking:1 uses the location from b5 and OffstreetParking:2 combines b4 and b5

6.3.4.2 Query Entities Of Type OffstreetParking Via POST

TP Id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_04_02
Test objective	Five brokers are set up b1, b2, b3, b4 and b5. b1 has two registrations, one auxiliary for the entity in b2, one inclusive for the entity in b3. b2 has two registrations, one redirect for the entity in b4 and one redirect for the entity in b5. b3 has two exclusive registrations to b5. Check that the OffStreetParking:1 entity has the same location attribute as the one in b5. Check that the OffStreetParking:2 entity in b1 has the same availableSpotsNumber and totalSpotsNumber as the ones found in b4. Check that the OffStreetParking:2 entity in b1 has the same location attribute as the one in b5.
Reference	ETSI GS CIM 009 V1.8.1 [1], clauses 4.3.3, 4.3.6, 5.7.2, 6.23.2.1
Config Id	IOP_CNF_04
PICS Selection	PICS_DISTRIBUTED and PICS_QueEnt_AND_QueEntPos
Pre Conditions	• **Context:** No user context used. • **Data:** Data in following brokers: – b4 contains OffStreetParking1 with location and name only (Figure 4.1.1.1-3). – b4 contains Full version of OffStreetParking2 (Figure 4.1.1.1-2). – b5 contains Full version of OffStreetParking1 (Figure 4.1.1.1-1). – b5 contains OffStreetParking2 with location and name only (Figure 4.1.1.1-5).
Registrations Established	• Auxiliary b1 to b2 (Figure 4.1.3.2-2). • Inclusive b1 to b3 (Figure 4.1.3.1-1). • Redirect b2 to b4 (Figure 4.1.3.4-1). • Redirect b2 to b5 (Figure 4.1.3.4-2). • Exclusive b3 to b5 (Figure 4.1.3.3-2). • Exclusive b3 to b5 (Figure 4.1.3.3-3).
TP Permutation id	IOP_TP/NGSI-LD/Interoperability/Cons/E/IOP_003_04_02
Name	IOP_003_04_02 Query Entities Of Type OffstreetParking Via POST
Parent Release	v1.6.1
Expected behaviour	**Test events**
	1: Client sends an HTTP POST request to b1 to retrieve Entities with type OffstreetParking **2:** Agent checks that a success response has been returned with OffstreetParking:1 containing "location" and OffstreetParking:2 containing "availableSpotsNumber", "totalSpotsNumber" and "location" **3:** Client sends an HTTP POST request to b4 to retrieve Entities with type OffstreetParking **4:** Client sends an HTTP POST request to b5 to retrieve Entities with type OffstreetParking **5:** Agent checks that OffstreetParking:1 uses the location from b5 and OffstreetParking:2 combines b4 and b5

Annex A (informative): Change history

Date	Version	Information about changes
November 2025	V0.0.1	First Early Draft from former ISG CIM work item
February 2026	V0.1.1	Stable Draft candidate
July 2026	V0.1.2	Final Draft candidate

History

Version	Date	Status
V1.1.1	October 2026	Publication